

Chemistry, 2008 EC, entrance exam

Grade 12 · Chemistry · 2008 E.C.

Time allowed: 120 minutes

Questions: 68

Total marks: 68

Name _____ Grade _____ ID No. _____ Date _____

1. A drug decomposes by zero-order kinetics with a rate constant of $2 \text{ mgmL}^{-1} \text{ month}^{-1}$. If the initial concentration is 100 mgmL^{-1} , how long will it take for the drug to decompose by 10%? (1)
 A. 2 months B. 3 months
 C. 5 months D. 4 months
-
2. The value of K_{eq} for the following reaction is 0.25. $\text{SO}_2(\text{g}) + \text{NO}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) + \text{NO}(\text{g})$ What is the value of K_{eq} at the same temperature for the reaction below? $2\text{SO}_2(\text{g}) + 2\text{NO}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) + 2\text{NO}(\text{g})$ (1)
 A. 0.25 B. 0.062
 C. 0.50 D. 16
-
3. Which of the following would react to form pentyl ethanoate? (1)
 A. 1- propanol and pentanoic acid B. Ethanol and pentanoic acid
 C. 1- pentanol and ethanoic acid D. Ethanol and ethanoic acid
-
4. Which of the following is correct when 34495 is rounded to three significant figures? (1)
 A. 345 B. 34500
 C. 344 D. 3840
-
5. What is the first step of the scientific method? (1)
 A. Making observations B. Forming a hypothesis
 C. Performing an experiment D. Predicting the result of an experiment
-
6. What is the distance that a radio wave will travel in 0.250sec? (1)
 A. $1.2 \times 10^7 \text{m}$ B. $12 \times 10^7 \text{m}$
 C. $7.5 \times 10^7 \text{m}$ D. $5.6 \times 10^7 \text{m}$
-
7. Which of the following is not a form of chemical bonding? (1)
 A. Covalent bonding B. Metallic bonding
 C. Ionic bonding D. Hydrogen bonding
-
8. Which of the following statements is not true about covalent bonding? (1)
 A. Covalent compounds are least likely to be formed between atoms of the same elements.
 B. Covalent bonds are least likely to be formed between atoms of different elements on the right side of the periodic table.
 C. Covalent bonds are least likely to be formed between an element in Group I and an element in Group VII.
 D. Covalent bonds are least likely to be formed by head of the group elements with high ionization energies.
-
9. If we increase the concentration of a reactant, what happens to the collision between particles? (1)
 A. There are more collisions.
 B. There are fewer collisions.
 C. There are the same number of collisions, but they have less energy
 D. There are the same number of collisions, but they have more energy
-
10. What mass of magnesium is plated out upon electrolysis from molten MgCl_2 using a current of 60A for a period of 4000 seconds? (1)
 A. 30gm B. 24gm
 C. 60gm D. 72gm

11. For a voltaic (or galvanic) cell using Ag, Ag^+ (1.0M) and Zn, Zn^{2+} (1.0M) half cells, which of the following statements is incorrect? (1)
- The zinc electrode is the Anode.
 - The electrons will flow through the external circuit from the zinc electrode to the silver electrode.
 - The mass of the zinc electrode will decrease as the cell operates.
 - Reduction occurs at the zinc electrode as the cell operates.
12. The difference between fats and oils is that (1)
- Oils are liquid at room temperature.
 - Oils have more calories
 - Oils are solid at room temperature.
 - Fats are liquid at room temperature.
13. What is the solvent in 70% alcohol solution? (1)
- Water
 - Alcohol
 - Sugar
 - Kerosene
14. How many moles of H_2SO_4 are needed to prepare 5.0 liters of a 2.0M solution of H_2SO_4 ? (1)
- 2.5
 - 5
 - 20
 - 10
15. What is the balanced net ionic equation for the reaction of CaCl_2 and AgNO_3 ? (1)
- $\text{CaCl}_2 (\text{aq}) + 2\text{AgNO}_3 (\text{aq}) \rightarrow 2(\text{NO}_3)_2 (\text{aq}) + 2\text{AgCl} (\text{s})$
 - $\text{Ca}^{2+} (\text{aq}) + 2\text{Cl}^- (\text{aq}) + 2\text{NO}_3^- (\text{aq}) \rightarrow \text{Ca}^{2+} (\text{aq}) + 2\text{NO}_3^-$
 - $\text{Cl}^- (\text{aq}) + \text{Ag}^+ (\text{aq}) \rightarrow 2\text{AgCl} (\text{s})$
 - $2\text{Cl}^- (\text{aq}) + 2\text{Ag}^+ (\text{aq}) \rightarrow 2\text{AgCl} (\text{s})$
16. The K_a of Hypochlorous acid (HClO) is 3.0×10^{-4} at 25°C . What is the % ionization of hypochlorous acid in a 0.015M aqueous solution of HClO at 25°C ? (1)
- 2.1×10^{-5}
 - 0.14
 - 1.4×10^{-3}
 - 3.3×10^{-3}
17. Which of the following combinations cannot produce a buffer solution? (1)
- HClO_4 and NaClO_4
 - HCN and NaCN
 - HNO_2 and NaNO_2
 - NH_3 and $(\text{NH}_4)_2\text{SO}_4$
18. What is the molality of a solution that contains 51.2gm of naphthalene, C_{10}H_8 in 500ml of carbon tetrachloride? The density of CCl_4 is 1.60gm/ml (1)
- 0.750m
 - 0.500m
 - 0.84m
 - 1.69m
19. What is the ionization constant for a weak acid, H_a , that is 1.60% ionized in 0.0950M solution? (1)
- 2.69×10^{-3}
 - 3.77×10^{-2}
 - 2.47×10^{-5}
 - 9.91×10^{-6}
20. Which of the following is correct? (1)
- $1\text{pa} = 10\text{Nm}^2$
 - $1\text{N} = 10\text{Kg}$
 - $0.00072 = 7.2 \times 10^{-3}$
 - $1\text{L} = 1\text{dm}^3$
21. For the Galvanic cell shown below, which one of the following statement is correct? (1)
- [Figure: A hand-drawn galvanic cell. Two beakers stand side by side, each holding a solution drawn as horizontal wavy lines with a faint formula written inside the liquid. A metal strip dips into each beaker: the left strip is labelled "Magnesium" by a leader line coming in from the left, the right strip is labelled "Zinc" by a leader line coming in from the right. A wire runs up from each strip and the two wires meet at a small circle containing the letter "V" (a voltmeter) at the top centre, closing the external circuit. Beneath that wire, a second inverted U-tube joins the two beakers, dipping into both solutions; the word "Salt" is printed in the gap between the beakers, with a short tick mark on the tube above it, labelling it as the salt bridge. No numerical values are shown.]*
- At the zinc electrode, zinc ions are formed.
 - The electrode potential is measured by the voltmeter.
 - The following reaction takes place at the magnesium electrode $\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}$.
 - Electrons flow from the zinc electrode to the magnesium electrode.

22. What is the most common Ore used for the extraction of copper? (1)
- A. CuO B. CuSO_4
C. CuCO_3 D. CuFeS_2
23. Which of the following statements about polyvinyl chloride is not correct? (1)
- A. Pvc can be used in making water pipes B. Pvc is stiff
C. Pvc is softened on heating D. The monomer of Pvc is $\text{CHCl} = \text{CHCl}$
24. Which of the following is a natural polymer? (1)
- A. Keratin B. Polythene
C. Cellulose D. Polymethyl methacrylate
25. Which of the following electrons, identified only by their n and l quantum numbers have the highest energy? $n=3, l=0$ $n=4, l=1$ $n=3, l=2$ $n=4, l=2$ (1)
- A. $n = 3, l = 2$ B. $n = 4, l = 1$
C. $n = 4, l = 2$ D. $n = 3, l = 0$
26. What is the maximum number of unpaired electron in a d subshell? (1)
- A. 2 B. 5
C. 3 D. 4
27. The following energy level diagram represents the out most shell of what ground state element? (1)
[Figure: A hand-drawn orbital (box) energy-level diagram of an outermost shell. At the lower left is a single short horizontal line with a small box sitting on it holding a pair of arrows drawn as "↑" and an up arrow together (a filled, paired orbital); the label "3s" is written in a box directly beneath that line. Higher up and to the right are three short horizontal lines of equal height, drawn side by side: the leftmost carries a small box holding a single arrow (one unpaired electron), and the middle and right lines are bare. The label "3p" is written in a box beneath the leftmost of those three lines. So the diagram shows a full 3s level below and three 3p levels above, of which only the first holds one electron.]
- A. B B. He
C. Al D. Be
28. In order to advance the level of a theory, a hypothesis should: (1)
- A. Be obviously accepted by most people.
B. Be repeatedly confirmed by experimentation.
C. Be a fully functional experiment.
D. Report the past experience.
29. Which of the following types of rays combine to form atoms of Helium? (1)
- A. Gamma (γ) rays B. Beta (β)
C. Alpha (α) D. X-rays
30. What is the relationship between frequency (ν), wavelength (λ) and the speed of light (c)? (1)
- A. $\lambda/\nu = c$ B. $\nu c = \lambda$
C. $hc/\lambda = \nu$ D. $c = \nu\lambda$
31. What is the magnitude of quantum energy and the frequency for an object whose wave length is $0.6 \times 10^{-6}\text{m}$? (1)
- A. $3.31 \times 10^{-9}\text{J}, 5 \times 10^{14}\text{s}^{-1}$ B. $3.98 \times 10^{-40}\text{J}, 2 \times 10^{-15}\text{s}^{-1}$
C. $1.99 \times 10^{-25}\text{J}, 3.98 \times 10^{-40}\text{s}^{-1}$ D. $9.94 \times 10^{-12}\text{J}, 1.99 \times 10^{-25}\text{sec}$
32. What new concept did Bohr adapt and use to formulate his made of the atom? (1)
- A. Electromagnetic theory developed by Maxwell.
B. The quantum concept developed by plank
C. Photoelectric theory developed by Thompson
D. Neutron theory developed by Chadwick.
33. What values of l are permitted for an electron with $n = 4$? (1)
- A. 1, 2, 3 B. 1, 2, 3, 4
C. 0, 1, 2, 3, 4 D. 0, 1, 2, 3

34. What would happen to the oxygen molecule upon ionization to O_2^+ ? (1)
- The bond length will increase and the bond energy will increase.
 - The bond length will increase and the bond energy will decrease.
 - The bond length will decrease and the bond energy will increase.
 - The bond length will decrease and the bond energy will decrease.
-
35. For a first order reaction, a plot of _____ versus _____ is linear. (1)
- $1/[A]_t$, t
 - $\ln 1/[A]_t$, t
 - $[A]_t$, t
 - $\ln [A]_t$, t
-
36. The rate law of the overall reaction $A + B \rightarrow C$ is $\text{rate} = K[A]^2$ which of the following will not increase the rate of the reaction? (1)
- Increasing the concentration of reactant A.
 - Increasing the concentration of reactant B.
 - Increasing the temperature of the reaction.
 - Adding a catalyst for the reaction.
-
37. How many bonding pairs and lone pairs, respectively does the ion ICl_4^- have? (1)
- 3, 2
 - 4, 2
 - 5, 1
 - 4, 1
-
38. Which of the following molecule does not have a tetrahedral central atom? (1)
- SF_4
 - AlH_4^-
 - BF_4^-
 - $SiCl_4$
-
39. Acrylonitrile has the following Lewis structure with the designation of x, y and z for each carbon atom. What will be the value of the bond angle and geometry of (1)
- [Figure: A hand-drawn structural formula in two rows, printed as one image. Top row: "CH2 = C - C ≡ N", i.e. a CH2 group joined by a double bond to a carbon, that carbon joined by a single bond to a further carbon, and that carbon joined by a triple bond to N. A single vertical bond drops from the middle carbon of the top row. Directly below that vertical bond is the letter H. Below the H is a second row reading "C - C ≡ N": a carbon joined by a single bond to a carbon, which is triple-bonded to N; the left carbon of this row carries the vertical bond up to the H above it, and a further vertical bond descending below it, which is clipped off at the bottom edge of the image. The x, y and z carbon labels named in the stem are not visible anywhere in the rendered picture.]*
- 1090, tetrahedral
 - 1200, trigonal pyramidal
 - 1800, linear
 - 900, T- shape
-
40. Antimony (Sb) is a group v element, what will be the molecular geometry and number of lone pair electrons, respectively that exists in the ion $[SbCl_5]^{2-}$? (1)
- Seesaw, 1
 - Square pyramid, 2
 - Seesaw, 2
 - Linear, 3
-
41. Which of the following molecules does not have a trigonal bipyramidal electron-pair geometry? (1)
- SF_4
 - ClF_3
 - XeF_2
 - BrF_5
-
42. How many atomic orbitals are required for an sp^3 hybridization? (1)
- 2
 - 6
 - 4
 - 8
-
43. Formic acid, which is released by ants, has a molecular formula of $HCOOH$. What are the possible hybridizations that exist in the molecule? (1)
- Sp^2 and sp^3
 - Sp and sp^3
 - Sp , sp^2 and sp^3
 - Sp and sp^2
-
44. Which of the following statements concerning fats and oils is incorrect? (1)
- They are also called triacylglycerols.
 - They are also called triglycerides.
 - They are fatty acid salts.
 - They are glycerol trimesters.

45. Which of the following statement(s) is (are) applicable to a balanced chemical equation of an elementary reaction? I. Order is same as molecularity. II. Order is less than the molecularity. III. Order is greater than the molecularity. IV. Molecularity can never be zero. (1)
- A. I
B. I, II
C. I, IV
D. I, III
-
46. At high pressure, the following reaction is zero order, Which of the following options is (are) correct for this reaction? I. Rate of reaction = rate constant. II. Rate of reaction depends on the concentration of ammonia. III. Rate of decomposition of ammonia remains constant until ammonia decomposes completely. IV. Further increase in pressure will change the rate of reaction. $2\text{NH}_3(\text{g}) \xrightarrow{1130\text{ K, pt}} \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ (1)
- A. I
B. I, III, IV
C. I, II
D. I, II, IV
-
47. Which of the following expressions is correct for the rate of reaction given below? (1)
- $$5\text{Br}^-(\text{aq}) + \text{BrO}_3^- + 6\text{H}^+(\text{aq}) \rightarrow 3\text{Br}_2(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$$
- A. $\frac{\Delta[\text{Br}]}{\Delta t} = \frac{5[\Delta H]}{\Delta t}$
B. $\frac{\Delta[\text{Br}]}{\Delta t} = 5/6 \frac{\Delta H}{\Delta t}$
C. $\frac{\Delta[\text{Br}]}{\Delta t} = 6/5 \frac{\Delta H}{\Delta t}$
D. $\frac{\Delta[\text{Br}]}{\Delta t} = 6 \frac{\Delta H}{\Delta t}$
-
48. The following equilibrium constants were determined at 300°C . $2\text{N}_2\text{O}(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + \text{O}_2(\text{g})$, $K_c = 4.0 \times 10^{18}$ (1)
 $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$, $K_c = 4.0 \times 10^{-31}$ What will be the equilibrium constant at 300°C for the gaseous reaction of
 $2\text{N}_2\text{O}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 4\text{NO}(\text{g})$?
- A. 3.2×10^{-12}
B. 6.4×10^{45}
C. 5.0×10^{50}
D. 1.6×10^{-49}
-
49. When 0.50 mol of N_2O_4 is placed in a 4.0 liter reaction vessel and heated to 400K , 80% of the N_2O_4 decomposes to NO_2 gas as follows: (1)
 $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ What will be the value of K_p , in units of pressure, at 400K for this reaction?
- A. 2.62
B. 13.12
C. 50.48
D. 16.2
-
50. Which of the following is not true about carbonyl compounds? (1)
- A. Carbonyl compounds contain 3σ - bond and 1π - bond.
B. The carbon oxygen bond is both longer and weaker.
C. The bond angle in carbonyl is about 120° .
D. Carbonyl compounds may be hydrolysed.
-
51. Sodium acetate spontaneously crystallizes out of a supersaturated solution on standing. Which of the following is true for the thermodynamic quantities of this system for such a process? (1)
- A. $\Delta S < 0$, $\Delta H < 0$
B. $\Delta G < 0$, $\Delta H > 0$
C. $\Delta S > 0$, $\Delta G < 0$
D. $\Delta S > 0$, $\Delta H > 0$
-
52. In which of the following systems will the position of equilibrium shift to the left upon an increase in pressure, but to the right upon an increase in temperature? (1)
- A. $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g})$, $\Delta H > 0$.
B. $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{C}_2\text{H}_5\text{OH}(\text{g})$, $\Delta H < 0$.
C. $\text{C}_2\text{H}_6(\text{g}) \rightleftharpoons \text{C}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g})$, $\Delta H > 0$.
D. $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $\Delta H > 0$.
-
53. The hydrogen used in the Haber process is made by the following reaction $\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + 3\text{H}_2(\text{g})$, $\Delta H^\circ = +206\text{KJ}$ Which of the following sets of conditions will favour the formation of H_2 ? (1)
- A. Low pressure and high temperature.
B. Low pressure and low temperature.
C. High pressure and low temperature.
D. High pressure and high temperature.
-
54. Which of the following types of compounds are expected products from the saponification of a fat? (1)
- A. Glycerol and fatty acid salts.
B. Glycerol and fatty acids.
C. Fatty acid salts and fatty acids.
D. Glycerol, fatty acid salts and fatty acids.

55. Which one of the following statements best describes the standard enthalpy of formation of any element? (1)
- The value of ΔH_f^0 (element) depends on temperature.
 - The value of ΔH_f^0 (element) is zero for any element in the standard state.
 - The value of ΔH_f^0 (element) is zero only for elements in the solid state.
 - The value of ΔH_f^0 (element) is zero only at absolute zero temperature.
56. Equal masses of He and Ne are placed in a sealed container. What is the partial pressure of Ne, if the total pressure is 6 atm? (1)
- 2
 - 3
 - 1
 - 5
57. What is the molarity of a solution made by dissolving 10g of glucose ($C_6H_{12}O_6$) in sufficient water to form 200ml solution? (1)
- 0.18
 - 0.251
 - 0.362
 - 0.278
58. In which direction will the following equilibrium shift, if a solution of CH_3CO_2Na is added? (1)
- $$CH_3COOH(aq) \rightleftharpoons CH_3CO_2^-(aq) + H^+(aq)$$
- The equilibrium shift to the right (more products).
 - No change
 - The equilibrium shifts to the left (more reactant)
 - Cannot be predicted.
59. A 1.0M solution of a weak acid is found to dissociate by only 1.37%. Which of the following acids is it likely to be? (1)
- HF $K_a = 7.2 \times 10^{-4}$
 - HNO_2 $K_a = 7.2 \times 10^{-4}$
 - CH_3COOH $K_a = 1.8 \times 10^{-5}$
 - $HCOOH$ $K_a = 1.9 \times 10^{-4}$
60. Consider the following graph, which relates to the equilibrium system: $CH_3COOH(aq) + H_2O \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$, $\Delta H < 0$ (1)
- Which of the following actions caused the change in the concentration of $[H_3O^+(aq)]$ at time t ?
- [Figure: A hand-drawn line graph with no numerical scale on either axis. The vertical axis, drawn with an arrowhead at the top, is labelled $[H_3O^+(aq)]$ written sideways up the axis. The horizontal axis, also arrow-headed, is labelled "Time" at its right end, with the letter "t" printed beneath it further left. A heavy vertical dashed line runs the full height of the plot at the position marking time t . The plotted curve begins at a constant, relatively high value on the vertical axis and runs perfectly flat and horizontal up to the dashed line. At the dashed line it turns downward, falling steeply at first and then curving over into a smooth decay, and finally flattens out at a new, lower constant value that continues horizontally to the right edge of the graph. So the concentration is steady, drops sharply at time t , and settles at a lower steady value.]
- Addition of $CH_3COO^-(aq)$
 - Addition of HCl
 - Decreasing of temperature.
 - Increasing the volume of the container.
61. A $1.0 \times 10^{-4}M$ solution has a pH of 10.00. The solution is a (1)
- Weak acid
 - Weak base
 - Strong base
 - Strong acid
62. A polysaccharide is a polymer made up of which kind of monomers? (1)
- Amino acids
 - Nucleotides
 - Simple sugars
 - Alternating sugar and phosphate groups.
63. What is the purpose of a salt bridge in an electrochemical cell? (1)
- To provide a source of ions to react at the anode and cathode.
 - To maintain electrical neutrality in the half-cell through migration of ions.
 - To provide means of electrons to travel from the cathode to the anode.
 - To provide means of electrons to travel from the anode to the cathode.
64. What is galvanized iron? (1)
- Iron that is coated with tin.
 - Iron that is coated with zinc.
 - Iron that is coated with chromium.
 - Iron that is coated with aluminium.

65. The standard cell potential (E^0) for the reaction below is 1.10v. What is the cell potential for this reaction when $[Cu^{2+}] = 1 \times 10^{-5}M$ and $[Zn^{2+}] = 1M$? $Zn_{(s)} + Cu_{(aq)}^{2+} \rightarrow Zn_{(aq)}^{2+} + Cu_{(s)}$ (1)
- A. 1.1
B. 0.95
C. 1.2
D. 1.35

66. Which of the following statements is true? (1)
- A. The more positive the value of E_{red}^0 , the greater the driving force for reduction.
B. The more exothermic the value of E_{red}^0 , the greater the driving force for reduction.
C. The more endothermic the value of E_{red}^0 , the greater the driving force for reduction.
D. The more negative the value of E_{red}^0 , the greater the driving force for reduction.

67. Which of the following types of elements are good oxidizing agents? (1)
- A. Alkali metals
B. Halogens
C. Lanthanides
D. Transition elements

68. Which of the following statements concerning petroleum is Not correct? (1)
- A. It is a renewable energy source.
B. It is a fossil fuel.
C. It is a mixture consisting mainly of hydrocarbons.
D. It was formed from marine organisms, which died millions of years ago.

— END OF EXAMINATION —